

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examination-2023 Course: B. Tech Branch: All Semester: I Course Code & Name: Engineering Mathematics-I (BTBS101) Max Marks: 60 Date:01-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Find the rank of matrix by converting it into Normal Form $A = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 2 & 2 & -1 & 1 \\ -1 & -1 & 1 & -1 \\ 2 & 1 & -1 & 2 \end{bmatrix}$	Understand (CO1)	6
B)	Find eigen values & eigen vector for largest eigen value for the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	Understand (CO1)	6
C)	Check the consistency and solve: $2x - 3y + 5z = 1, 3x + y - z = 2, x + 4y - 6z = 1$	Understand (CO1)	6
Q.2	Solve Any Two of the following.		12
A)	If $z(x+y) = x^2 + y^2$, show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$	Understand (CO2)	6
B)	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$	Understand (CO2)	6
C)	If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, then find the value of $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$	Understand (CO2)	6
Q.3	Solve Any Two of the following.		12
A)	If $u = x + 2y^2 - z^3, v = 2x^2yz, w = 2z^2 - xy$ then evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$	Understand (CO3)	6
B)	Discuss the maxima and minima for the function $x^2 + y^2 + (30 - x - y)^2$ and hence find the extreme value of the function.	Understand (CO3)	6
C)	Using Lagrange's undetermined multipliers find the maximum value of $x^2 + y^2 + z^2$ when $x + y + z = 3a$	Understand (CO3)	6

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Q.4	Solve Any Two of the following.		12
A)	Evaluate $\int_0^a x^3(a-x)^{\frac{3}{2}} dx$	Understand (CO4)	6
B)	Trace the curve $y^2(2a-x) = x^3$.	Understand (CO4)	6
C)	Trace the curve $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$.	Understand (CO4)	6
Q. 5	Solve Any Two of the following.		12
A)	Evaluate $\int_0^1 \int_0^{z^2} \int_0^{z^2-x} xz \, dx \, dy \, dz$	Understand (CO5)	6
B)	Find the area bounded by $y^2 = 4x$ and $2x - 3y = -4$.	Understand (CO5)	6
C)	Change to polar and evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx \, dy$.	Understand (CO5)	6